

Wantian Huang

Economist: empirical IO, demand estimation and pricing, causal inference and experimentation

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EDUCATION

Ph.D. Economics, University of Maryland, College Park

Expected May 2027

M.A. Economics, Columbia University

2020

B.S. Economics & Mathematics, University of California San Diego

2018

PROFESSIONAL EXPERIENCE

Economist Intern, Amazon Ads Marketing Finance

Seattle, WA Jun 2026 – Aug 2026

- Evaluated five randomized promotions in an advertising-credit program and showed that credits raised seller ad adoption without incremental ad revenue, with returns varying widely across segments; estimated intent-to-treat effects by covariate-adjusted OLS, with difference-in-differences and event studies for dynamics under repeated offers.
- Found that effects fade with repeated offers and concentrate among sellers with high predicted take-up (causal forests, compliance-score matching, double machine learning), supporting capped and targeted offers.
- Authored the technical review and briefed the Ads finance director; quantified millions of dollars in annual credits with no incremental return and recommended credit-menu and re-offer changes plus a follow-on experiment.

Research Assistant to Prof. Chenyu Yang, University of Maryland

Summers 2024 – 2025

- Built and analyzed large datasets (Nielsen RMS/HMS, streaming analytics) in Python, Stata, and MATLAB; implemented econometric and structural models of demand estimation, merger simulation, and counterfactual analysis.

RESEARCH

Testing the Substitution Predictions of Demand Models: Evidence from the Blue Bell Recall

Job Market Paper

- Use the 2015 Blue Bell recall, a whole-brand withdrawal, to test out of sample where a removed product's sales go; estimate logit, nested logit, and random-coefficient (BLP, RCNL) models on 3.7M product-market-months of Nielsen scanner data with Homescan micro-moments.
- Find that rivals recaptured 77% of Blue Bell's volume, half via private label; non-nested models retain under 50%, while nests pooling private label with Blue Bell's tubs retain 73–74% and best match which rivals gain.
- Show that product grouping, more than taste heterogeneity, drives predicted substitution and the value of a brand's availability (\$23M vs. \$34M for one summer), informing assortment, stockout, and merger analysis.

Economies of Scope in Retail Distribution: A Study of Ready-to-Consume Categories

with Ying Fan, Yidan Xu, and Chenyu Yang

- Show that economies of scope raise product variety and consumer welfare, and that ignoring them understates the gains from small-firm mergers.
- Estimate demand with a BLP random-coefficients model and micro-moments on Nielsen RMS scanner data, recovering elasticities, diversion ratios, and marginal costs for merger simulation.
- Bound fixed costs under minimal assumptions using moment inequalities with optimal-transport inference; run estimation and counterfactual pipelines on an HPC cluster.

Evaluating the Welfare Impacts of EV Charging Station Subsidies: An Analysis of the NEVI Program in Texas

- Assess cost-effectiveness of 50 subsidized stations: \$31.8M in fixed costs against \$15.2M in annual expected welfare gains, implying roughly a two-year payback.
- Estimate EV demand with structural models that accommodate prevalent zero market shares.

Quality Uncertainty, Learning, and Product Variety in Live Streaming Platforms

In progress

- Quantify the welfare contribution of marginal streamers and evaluate counterfactual monetization, retention, and merger scenarios.
- Develop a nested two-stage demand framework for viewer engagement; model streamer learning, effort, and exit.

Presentation: Industrial Organization Workshop, University of Maryland (scheduled, October 2026)

TEACHING

Instructor, University of Maryland: Economics of Regulation and Antitrust (Summer 2024)

Teaching Assistant, M.S. in Applied Economics (2021 – 2026): 29 course-terms across 11 graduate courses

TECHNICAL SKILLS

- Programming: Python (pandas, NumPy, scikit-learn, pyfixest, EconML), SQL (Redshift, Hive, Spark), Stata, MATLAB, LaTeX; Parquet panels, HPC (SLURM) and distributed computing
- Methods: A/B testing and RCTs, DiD (incl. staggered), event studies, IV/RD, causal forests, double machine learning; discrete-choice demand (logit, nested logit, BLP, RCNL), merger simulation, moment inequalities